

WARIN – Cost Analyzer

Functional Requirement Note

1. Objective

Add a new menu in WARIN:

After Performance Card → Cost Analyzer

Purpose of Cost Analyzer is to provide management visibility into:

- Total employee cost for the selected period
- Cost captured against Listed Projects
- Cost captured as Unplanned Work
- Cost not captured through WARIN work posting
- Over-captured work/cost exceptions
- Department-wise cost contribution
- Project-wise cost contribution
- Unplanned reason-wise cost contribution
- Employee-level contribution through drill-down

This module will calculate only **employee manpower cost**.

No other project costs such as:

- Travel
- Cloud
- Hardware
- Software licence
- Vendor cost
- Reimbursement
- Material cost
- Other operational expenses

will be included in V1.

2. Menu Placement

Add new menu:

Cost Analyzer

Position:

Immediately after Performance Card

Access shall follow the existing WARIN **Page Rights / Role Rights** mechanism.

No separate access-control framework is required.

3. Required Master Enhancement

3.1 Employee Cost (It is a new master, Employee added via Employee master will only be shown in this)

Maintain employee-level:

- Employee ID
- Employee Name
- Cost Per Minute
- Effective From Date
- Status

Example:

Employee Cost / Minute Effective From

EMP001 ₹8.50 01-Apr-2026

EMP001 ₹9.25 01-Jul-2026

Historical cost calculations must use the cost rate applicable on the actual work date.

Do not overwrite historical rate records.

Whenever cost changes, create a new effective-dated record.

4. Cost Analyzer Filters

Top section shall contain:

4.1 Period

Only week/month-based period selection is required.

Support period options consistent with WARIN reporting pattern, such as:

- This Week
- Previous Week
- This Month
- Previous Month
- Custom Weeks

Do not provide individual date selection.

4.2 Department

Department dropdown:

- All Departments
- Individual Department

Department selection shall recalculate the complete analyzer.

When a specific department is selected:

- Project cost = only employees belonging to that department
- Unplanned cost = only employees belonging to that department
- Total CTC = only employees belonging to that department
- Lost Cost = only that department
- Project Cards = department-specific contribution
- Employee drill-down = department-specific employees

Employee department shall be taken from Employee Master.

Historical department movement logic is not required.

5. Core Cost Logic

5.1 Actual Captured Work

Only following work is treated as captured cost:

Captured Work = Listed Project Actual Work + Unplanned Actual Work

Exclude:

- Office Time
 - Lunch
 - Idle Time
 - Attendance duration
 - Non-posted time
 - Leave hours
 - Holiday hours
 - Other presence-related hours
-

6. Project Cost Calculation

For every project work entry:

Project Cost = Actual Project Minutes × Applicable Employee Cost Per Minute

Example:

Employee cost = ₹10/minute

Project actual = 180 minutes

Project Cost:

180 × ₹10 = ₹1,800

Project cost must always use:

- actual posted minutes

- employee cost applicable on work date

7. Unplanned Cost Calculation

For every Unplanned Work entry:

Unplanned Cost = Actual Unplanned Minutes × Applicable Employee Cost Per Minute

Example:

Employee cost = ₹10/minute

Unplanned work = 60 minutes

Unplanned Cost:

₹600

The existing Unplanned Reason shall also be retained for cost analysis.

8. Total CTC Cost Logic

Total CTC represents theoretical employee manpower cost for the selected period.

Do not derive Total CTC from attendance or leave.

Use configured standard daily working hours from existing WARIN master.

Saturday and Sunday must always be excluded.

Formula:

Employee Total CTC Cost

= Eligible Working Minutes × Applicable Cost Per Minute

Where:

Eligible Working Minutes

= Number of Monday-Friday days in selected period × Configured Working Hours Per Day × 60

Example:

Configured work hours/day = 8 hours

Week contains 5 weekdays:

$5 \times 8 \times 60 = 2,400$ minutes

Employee Cost = ₹10/minute

Total CTC:

$2,400 \times ₹10 = ₹24,000$

9. Leave / Holiday Treatment

For Total CTC calculation:

Ignore:

- Personal leave
- Public holiday
- Paid leave
- Unpaid leave
- Attendance absence

If Monday-Friday falls inside selected period, configured daily hours shall be counted.

Only Saturday and Sunday are automatically excluded.

10. Employee Joining / Exit Logic

Employee Total CTC should only be counted during active employment.

Use:

- Joining Date
- Exit / Relieving Date

Example:

Selected month = September

Employee joining date = 15 September

Calculate Total CTC only from:

15 September onward

Similarly, if employee exits on 20 September:
calculate only until 20 September.
Saturday/Sunday exclusion shall still apply.

11. Lost / Not-Captured Cost

Formula:

Raw Lost Cost

= Total CTC – Project Cost – Unplanned Cost

Normally:

Lost Cost = Total CTC – Captured Cost

Where:

Captured Cost = Project Cost + Unplanned Cost

12. Over-Captured Cost Exception

Captured Actual Work may exceed theoretical Total CTC.

Example:

Standard Weekly Capacity = 40 hours

Actual Project + Unplanned Work = 46 hours

Do not show negative Lost Cost.

Formula:

Lost Cost = MAX(0, Total CTC – Captured Cost)

And:

Over-Captured Cost

= MAX(0, Captured Cost – Total CTC)

Also calculate:

Over-Captured Hours

= MAX(0, Captured Actual Hours – Standard Eligible Hours)

Over-Captured Cost / Hours shall be shown separately as an exception.

Do not modify or cap actual posted work.

Actual posted work remains authoritative.

13. Top KPI Summary

Before the analytical charts, show summary KPI cards.

Recommended cards:

13.1 Total CTC

Show:

- ₹ Amount
 - Previous comparable period %
 - Increase / decrease indicator
-

13.2 Captured Cost

Show:

Project Cost + Unplanned Cost

Also show:

- ₹ Amount
 - % of Total CTC
 - Previous comparable period %
-

13.3 Project Cost

Show:

- ₹ Amount
- % of Total CTC

- Previous comparable period %

13.4 Unplanned Cost

Show:

- ₹ Amount
- % of Total CTC
- Previous comparable period %

13.5 Lost / Not-Captured Cost

Show:

- ₹ Amount
- % of Total CTC
- Previous comparable period %

13.6 Over-Captured Cost

Show only if applicable.

Show:

- ₹ Amount
- Hours
- Number of affected employees

This is an exception indicator.

14. Previous Period Comparison

Previous-period comparison is required only on the top-level KPI cards.

Do not show previous-period comparison throughout every chart/card.

Comparison must use equivalent period.

Examples:

This Week → Previous Week

Previous Week → Week before Previous Week

This Month → Previous Month

4 Custom Weeks → immediately preceding 4 comparable weeks

Show:

- Previous amount
- Percentage increase/decrease
- Direction indicator

15. Analyzer Part 1 – Project vs Unplanned Cost

Display a donut chart with two main components:

Segment 1

Listed Projects

Show:

- Cost amount
- % share of Captured Cost

Segment 2

Unplanned Work

Show:

- Cost amount
- % share of Captured Cost

Formula:

Project %

= Project Cost ÷ Captured Cost × 100

Unplanned %

$$= \text{Unplanned Cost} \div \text{Captured Cost} \times 100$$

Project + Unplanned must always equal:

100% of Captured Cost

16. Part 1 Drill-Down

Clicking Listed Projects

Open drill-down showing Project-wise breakdown.

For each project:

- Project Name
- Project Cost
- % share of Project Cost
- Actual Hours

Example:

Project	Cost	%	Hours
Project A	₹1,20,000	45%	210
Project B	₹90,000	34%	160
Project C	₹56,000	21%	104

Clicking Unplanned Work

Show Unplanned Reason-wise breakdown.

Example:

- Internal Meeting
- Training
- Support Activity
- Internal Discussion
- Administrative Work
- Other configured reasons

For each reason show:

- Reason
- Cost
- % share
- Hours

17. Analyzer Part 2 – Department Cost Distribution

Display donut chart showing department-wise Captured Cost.

Each segment represents:

Department Project Cost + Department Unplanned Cost

Show:

- Department Name
- Amount
- % contribution

Formula:

Department Cost Share

$$= \text{Department Captured Cost} \div \text{Company Captured Cost} \times 100$$

When analyzer itself is filtered to one department, this chart may either:

- show 100% selected department

or preferably:

- automatically switch to internal breakup view

Recommended V1 behaviour:

When specific Department is selected, replace department donut with:

Selected Department – Project vs Unplanned Cost

to avoid displaying a meaningless 100% donut.

18. Department Drill-Down

Clicking a Department shall open a side drawer.

Drawer shall show:

A. Project-wise Cost

For each project:

- Project Name
- Hours
- Cost
- % share within department

B. Unplanned Reason-wise Cost

For each reason:

- Unplanned Reason
- Hours
- Cost
- % share within department

C. Employee Contribution

Allow deeper drill-down to:

Department → Project/Unplanned → Employee

Employee rows show:

- Employee Name
 - Actual Hours
 - Cost
 - % contribution
-

19. Analyzer Part 3 – Project Cost Cards

Generate one card for every project where:

Selected Period Project Cost > ₹0

Project status does not matter.

Include project even if:

- Project started earlier
 - Project ended earlier
 - Project is currently closed
 - Work has been posted outside scheduled project dates
-

20. Project Card Fields

Each Project Card shall contain:

20.1 Project Name

Display project name prominently.

20.2 Project Dates

Show:

Start Date – End Date

20.3 PO Number

Show existing PO Number from Project Master.

If no PO exists, show:

—

20.4 Total Project Cost

Show:

₹ Cost incurred within selected analyzer period only

No lifetime project cost shall be shown.

20.5 Project Actual Hours

Recommended supporting value:

Actual Hours posted during selected period

20.6 Department-Wise Cost Contribution

Example:

Development

₹1,25,000 — 55%

QA

₹52,000 — 23%

CS

₹30,000 — 13%

Implementation

₹20,000 — 9%

Percentages are calculated within the project.

Formula:

Department Project Share

= Department Cost on Project ÷ Project Total Cost × 100

20.7 Company Cost Share

Show:

Project Cost Share in Company

Formula:

Project Cost in Selected Period

÷ Total Company Captured Cost in Selected Period × 100

Example:

Project Cost = ₹2,00,000

Company Captured Cost = ₹10,00,000

Project Company Share:

20%

Display:

₹2,00,000 | 20% of Company Captured Cost

21. Work Outside Project Dates

Cost posted before Project Start Date or after Project End Date must still be included in Project Cost.

Additionally calculate:

Outside Project Period Cost

Show on project card only when > ₹0.

Example:

Outside Project Period

₹32,000 | 13.3% of Project Cost

Also show corresponding:

- Outside Period Hours

This is informational and must not alter cost.

22. Project Card Drill-Down

Clicking a Project Card shall open side drawer.

Drawer structure:

A. Summary

- Project Name
 - Project Dates
 - PO Number
 - Selected Period
 - Actual Hours
 - Total Cost
 - Company Cost Share
 - Outside Project Period Cost
 - Outside Project Period Hours
-

B. Department Contribution

Columns:

- Department
 - Hours
 - Cost
 - % of Project Cost
-

C. Employee Contribution

Columns:

- Employee
- Department
- Actual Project Hours
- Cost
- % of Project Cost

This provides management visibility into:

Which employees are driving project cost

No individual employee details are required on the main dashboard.

23. Lost Cost Drill-Down

Clicking Lost / Not-Captured Cost KPI shall open a side drawer.

Show:

Employee Department Total CTC Captured Cost Lost Cost Capture %

Where:

Capture %

= Captured Cost ÷ Total CTC × 100

Also show:

- Standard Eligible Hours
- Actual Project Hours
- Actual Unplanned Hours
- Total Captured Hours
- Lost Equivalent Hours

Formula:

Lost Equivalent Hours

= Lost Cost ÷ Cost Per Hour

Use this only for analytical representation.

24. Over-Captured Drill-Down

Clicking Over-Captured KPI shall show affected employees.

Columns:

- Employee
- Department

- Standard Eligible Hours
- Actual Project Hours
- Actual Unplanned Hours
- Total Captured Hours
- Over-Captured Hours
- Total CTC
- Captured Cost
- Over-Captured Cost

This is an exception report.

Do not classify Over-Captured Cost as negative Lost Cost.

25. Important Percentage Denominators

Developer must maintain clear denominator logic.

Project vs Unplanned Donut

Denominator:

Captured Cost

Department Distribution

Denominator:

Company Captured Cost

Department inside Project

Denominator:

Project Cost

Employee inside Project

Denominator:

Project Cost

Project Company Share

Denominator:

Company Captured Cost

Lost Cost %

Denominator:

Total CTC

Capture %

Denominator:

Total CTC

26. Selected Period Rule

Every monetary figure displayed inside Cost Analyzer must relate only to the selected analyzer period.

This applies to:

- Total CTC
- Project Cost
- Unplanned Cost
- Lost Cost
- Over-Captured Cost
- Department Cost
- Project Cards
- Employee contribution

- Outside Project Period Cost

Do not show lifetime/project-to-date financial figures in V1.

27. Cost Calculation Date Principle

All cost calculation must be transaction-date based.

For each actual work entry:

1. Identify employee
2. Identify actual work date
3. Retrieve employee Cost Per Minute effective on that date
4. Retrieve actual work minutes
5. Calculate cost

This prevents future salary/cost changes from modifying historical analytics.

28. Recommended Main Screen Layout

Header

Cost Analyzer

Filters:

Period | Department

Row 1 – KPI Cards

- Total CTC
 - Captured Cost
 - Project Cost
 - Unplanned Cost
 - Lost Cost
 - Over-Captured Exception
-

Row 2

Left

Captured Cost Composition

Donut:

- Listed Projects
- Unplanned Work

Right

Department Cost Distribution

Donut:

- Department-wise Captured Cost
-

Row 3

Heading:

Project Cost Analysis

Responsive card grid containing all qualifying projects.

Provide sorting options:

- Highest Cost
- Lowest Cost
- Highest Company Share
- Highest Outside-Period Cost

Default:

Highest Cost First

29. No Separate Cost Database Snapshot Required for V1

Cost Analyzer should preferably calculate from existing operational records:

- Employee Master
- Employee Cost Master
- Project Master
- Project Actual Work Entries
- Unplanned Work Entries
- Department Master
- Working Hours configuration

Historical employee cost rates must, however, be persisted using effective dates.

Aggregation/caching can be introduced later if performance requires it.

30. Core Business Equations

Captured Cost

Captured Cost = Project Cost + Unplanned Cost

Lost Cost

Lost Cost = MAX(0, Total CTC – Captured Cost)

Over-Captured Cost

Over-Captured Cost = MAX(0, Captured Cost – Total CTC)

Therefore:

For normal case:

Total CTC = Captured Cost + Lost Cost

For over-captured case:

Captured Cost = Total CTC + Over-Captured Cost

Do not force:

Captured Cost ≤ Total CTC

because WARIN actual work posting must remain factual.

31. Example

Assume selected month:

Total CTC

₹12,00,000

Project Cost

₹7,20,000

Unplanned Cost

₹1,80,000

Then:

Captured Cost = ₹9,00,000

Lost Cost = ₹3,00,000

Capture %:

75%

Lost %:

25%

Captured Cost composition:

- Projects = 80%
- Unplanned = 20%

If Project Alpha Cost = ₹1,80,000:

Company Cost Share:

₹1,80,000 ÷ ₹9,00,000 = 20%

Therefore Project Card shows:

₹1,80,000 | 20% of Company Captured Cost

32. Functional Philosophy

Cost Analyzer is not intended to become an accounting or payroll system.

Its purpose is:

Convert WARIN work activity into manpower cost visibility.

The analytical hierarchy should remain:

Company

→ **Department**

→ **Project / Unplanned**

→ **Employee**

This gives management the ability to answer:

- How much manpower cost are we carrying?
- How much of it is captured through actual work?
- How much is going into projects?
- How much is going into unplanned work?
- How much cost is not represented through actual work posting?
- Which departments drive cost?
- Which projects consume the highest manpower cost?
- Which employees contribute most to a particular project's cost?
- Which projects continue consuming effort outside planned dates?
- Which employees are posting work beyond standard capacity?

The module should remain analytical and drill-down driven rather than becoming another operational data-entry screen.

33. Management Exception & Attention Layer

Cost Analyzer must not function only as a passive reporting dashboard. It shall automatically surface cost patterns requiring management attention.

33.1 Exception Indicators

Show a compact **Management Attention** section above the detailed charts/cards.

System shall identify and highlight:

- **Top Cost-Consuming Projects**
Projects contributing the highest share of Captured Cost in the selected period.
- **High Unplanned Cost**
Departments / employees / reasons where Unplanned Cost contribution is materially high.
- **High Lost / Not-Captured Cost**
Departments / employees with low Capture % and high Lost Cost.
- **Outside Project Period Cost**
Projects consuming manpower cost before Start Date or after End Date.
- **Over-Captured Work**
Employees where Captured Hours exceed standard eligible hours.

33.2 Initial V1 Approach

Do not introduce complex AI or statistical anomaly logic in V1.

Use simple ranking and threshold-based highlighting.

Examples:

- Top 5 projects by cost
- Top 5 departments/employees by Lost Cost
- Top 5 Unplanned Reasons by cost
- Projects with Outside Project Period Cost > 0
- Employees with Over-Captured Hours > 0

33.3 Exception Card Behaviour

Each exception should show:

- Exception Type
- Entity Name
- ₹ Cost / Hours
- Relevant %

- Short reason
- Click-through to existing drill-down

Example:

High Lost Cost

Development Department

₹1.85L Lost | 29% of CTC

→ View Department

33.4 Objective

The first management view of Cost Analyzer should answer:

“Where should I look first?”

Detailed donuts and project cards shall answer:

“Why is this happening?”

This distinction must be maintained in the UI design.